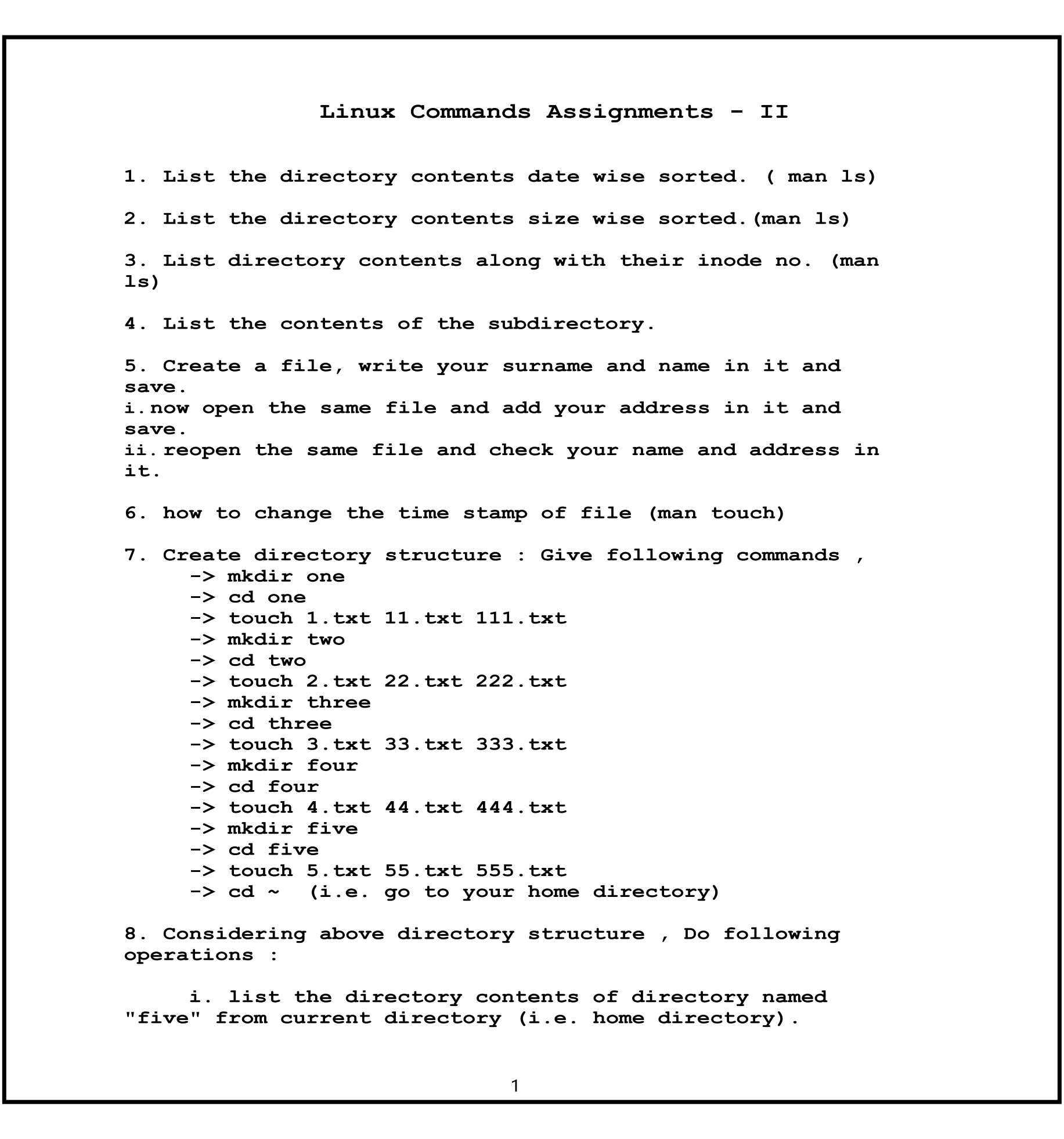
Assignment No 1



Linux Commands Assignments - II

1. List the directory contents date wise sorted. (man ls)

2. List the directory contents size wise sorted.(man ls)

3. List directory contents along with their inode no. (man ls)

4. List the contents of the subdirectory.

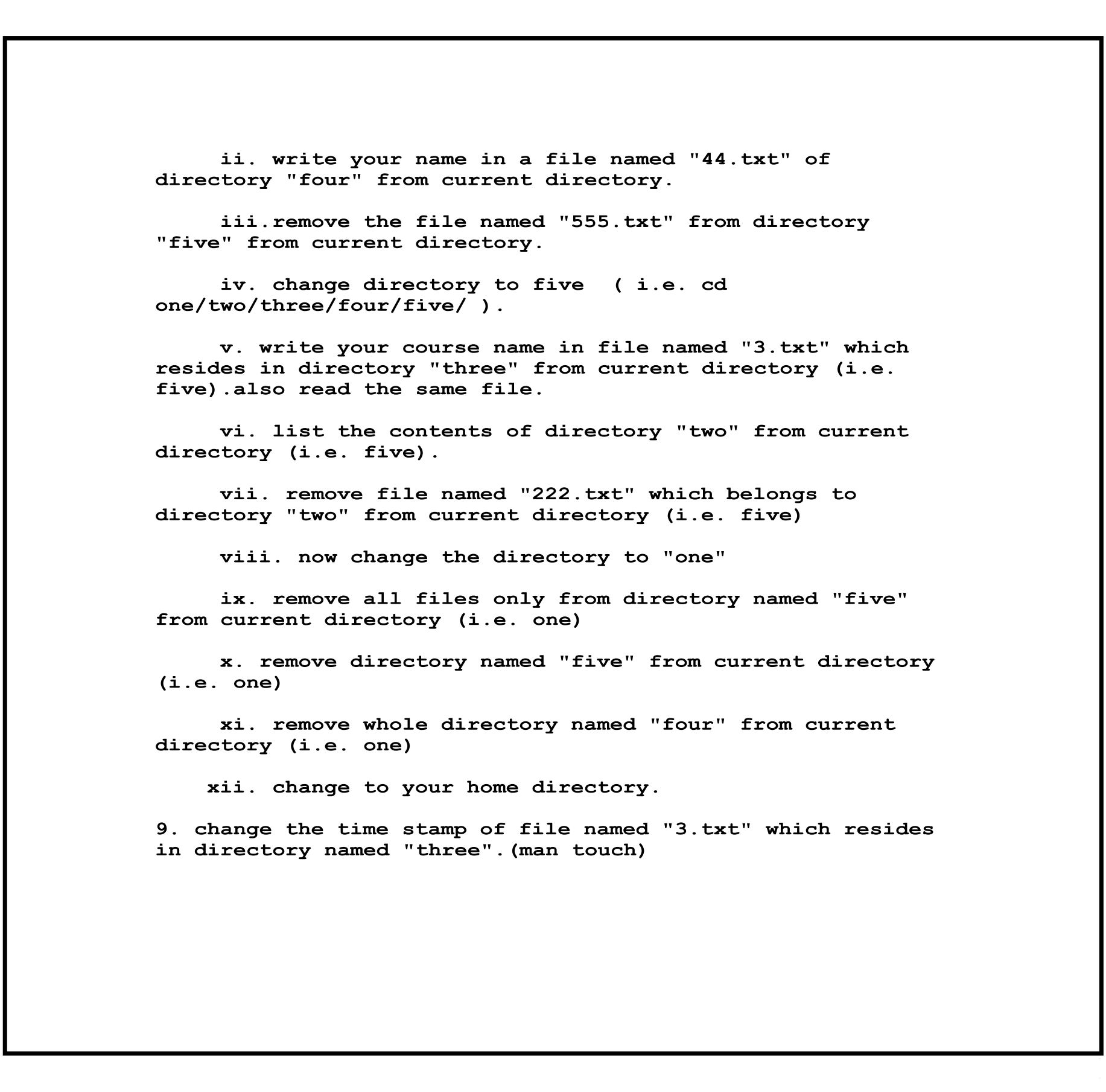
5. Create a file, write your surname and name in it and save.
i.now open the same file and add your address in it and save.
ii.reopen the same file and check your name and address in it.

6. how to change the time stamp of file (man touch)

7. Create directory structure : Give following commands ,
```
    -> mkdir one
    -> cd one
    -> touch 1.txt 11.txt 111.txt
    -> mkdir two
    -> cd two
    -> touch 2.txt 22.txt 222.txt
    -> mkdir three
    -> cd three
    -> touch 3.txt 33.txt 333.txt
    -> mkdir four
    -> cd four
    -> touch 4.txt 44.txt 444.txt
    -> mkdir five
    -> cd five
    -> touch 5.txt 55.txt 555.txt
    -> cd ~   (i.e. go to your home directory)
```

8. Considering above directory structure , Do following operations :

i. list the directory contents of directory named "five" from current directory (i.e. home directory).



ii. write your name in a file named "44.txt" of directory "four" from current directory.

iii.remove the file named "555.txt" from directory "five" from current directory.

iv. change directory to five (i.e. cd one/two/three/four/five/).

v. write your course name in file named "3.txt" which resides in directory "three" from current directory (i.e. five).also read the same file.

vi. list the contents of directory "two" from current directory (i.e. five).

vii. remove file named "222.txt" which belongs to directory "two" from current directory (i.e. five)

viii. now change the directory to "one"

ix. remove all files only from directory named "five" from current directory (i.e. one)

x. remove directory named "five" from current directory (i.e. one)

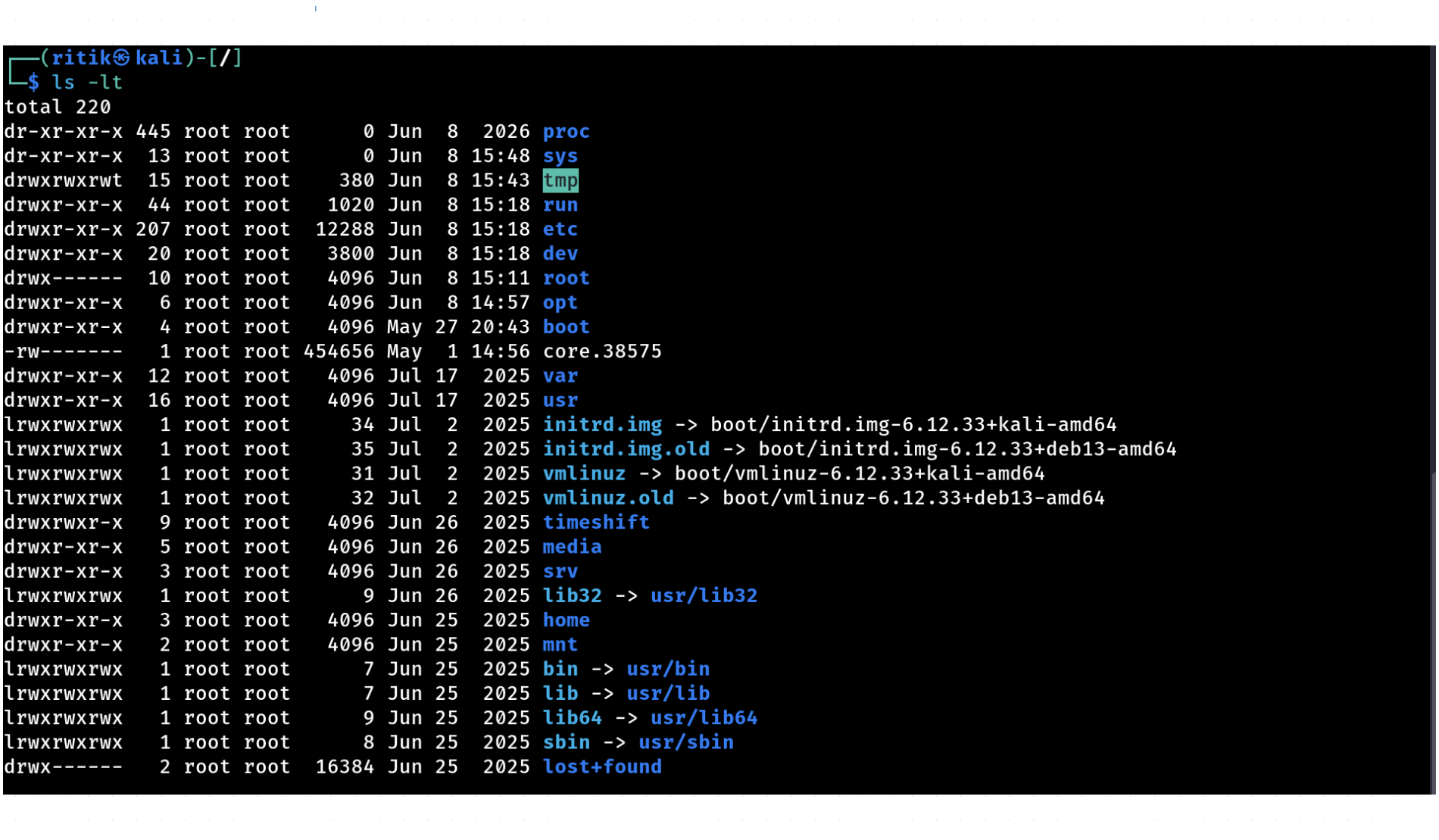
xi. remove whole directory named "four" from current directory (i.e. one)

xii. change to your home directory.

9. change the time stamp of file named "3.txt" which resides in directory named "three".(man touch)

1. List the directory contents date wise sorted. (man ls)

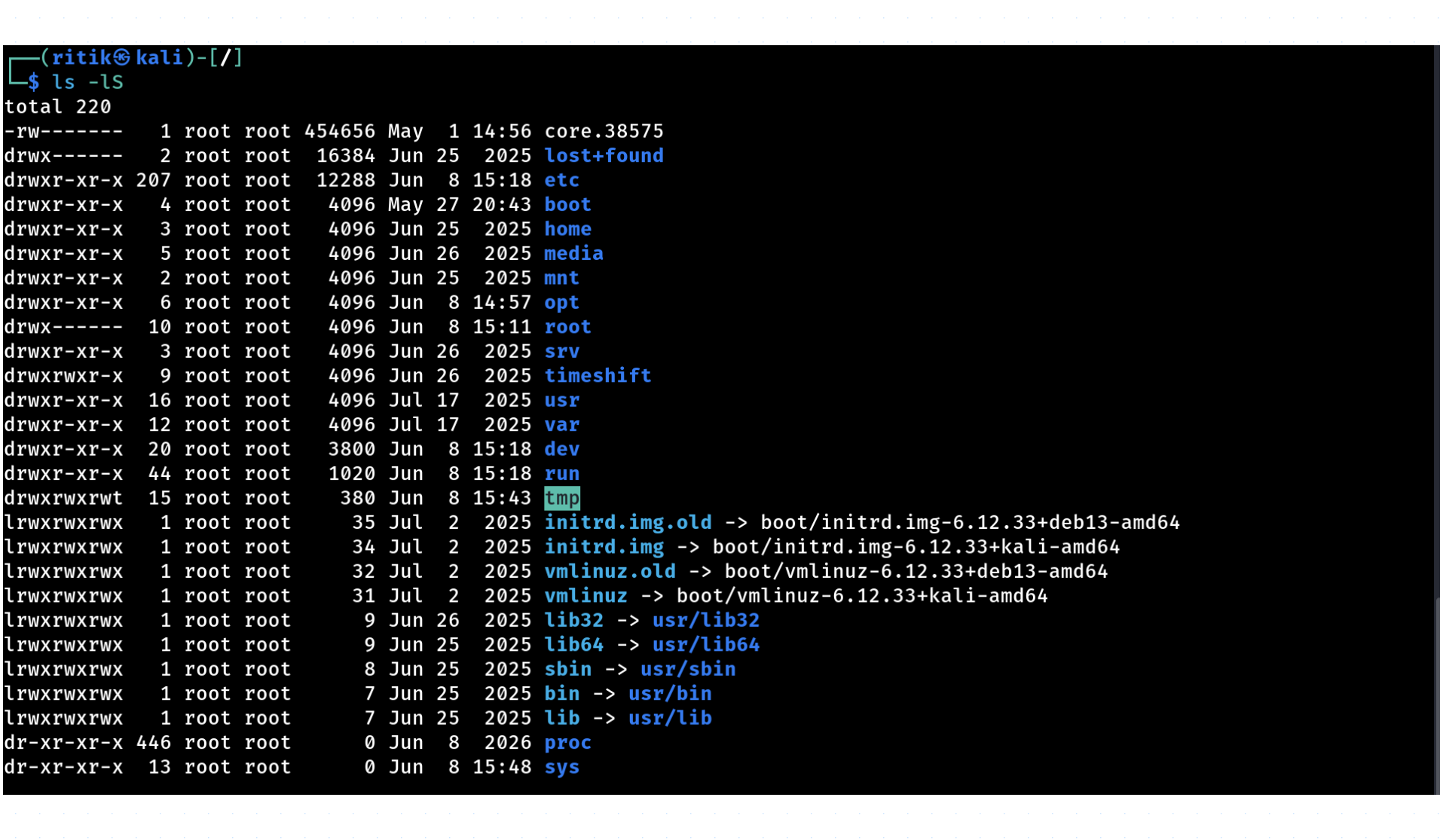
ls -lt
ls -ltr
ls --sort=time
ls -l --time=mtime
ls -lth



```
┌──(ritik㉿kali)-[/]
└─$ ls -lt
total 220
dr-xr-xr-x 445 root root      0 Jun  8  2026 proc
dr-xr-xr-x  13 root root      0 Jun  8 15:48 sys
drwxrwxrwt  15 root root    380 Jun  8 15:43 tmp
drwxr-xr-x  44 root root   1020 Jun  8 15:18 run
drwxr-xr-x 207 root root  12288 Jun  8 15:18 etc
drwxr-xr-x  20 root root   3800 Jun  8 15:18 dev
drwx------  10 root root   4096 Jun  8 15:11 root
drwxr-xr-x   6 root root   4096 Jun  8 14:57 opt
drwxr-xr-x   4 root root   4096 May 27 20:43 boot
-rw-------   1 root root 454656 May  1 14:56 core.38575
drwxr-xr-x  12 root root   4096 Jul 17  2025 var
drwxr-xr-x  16 root root   4096 Jul 17  2025 usr
lrwxrwxrwx   1 root root     34 Jul  2  2025 initrd.img -> boot/initrd.img-6.12.33+kali-amd64
lrwxrwxrwx   1 root root     35 Jul  2  2025 initrd.img.old -> boot/initrd.img-6.12.33+deb13-amd64
lrwxrwxrwx   1 root root     31 Jul  2  2025 vmlinuz -> boot/vmlinuz-6.12.33+kali-amd64
lrwxrwxrwx   1 root root     32 Jul  2  2025 vmlinuz.old -> boot/vmlinuz-6.12.33+deb13-amd64
drwxrwxr-x   9 root root   4096 Jun 26  2025 timeshift
drwxr-xr-x   5 root root   4096 Jun 26  2025 media
drwxr-xr-x   3 root root   4096 Jun 26  2025 srv
lrwxrwxrwx   1 root root      9 Jun 26  2025 lib32 -> usr/lib32
drwxr-xr-x   3 root root   4096 Jun 25  2025 home
drwxr-xr-x   2 root root   4096 Jun 25  2025 mnt
lrwxrwxrwx   1 root root      7 Jun 25  2025 bin -> usr/bin
lrwxrwxrwx   1 root root      7 Jun 25  2025 lib -> usr/lib
lrwxrwxrwx   1 root root      9 Jun 25  2025 lib64 -> usr/lib64
lrwxrwxrwx   1 root root      8 Jun 25  2025 sbin -> usr/sbin
drwx------   2 root root  16384 Jun 25  2025 lost+found
```

2. List the directory contents size wise sorted.(man ls)

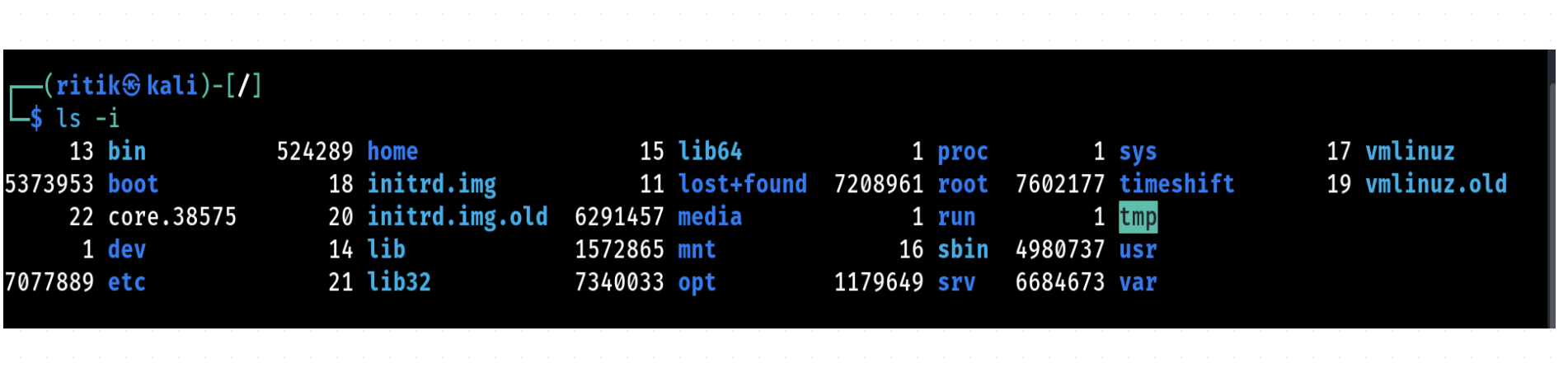
ls -lS
ls -lSr
ls -lhS
ls -lS --reverse
ls -lSh



```
┌──(ritik㉿kali)-[/]
└─$ ls -lS
total 220
-rw-------   1 root root 454656 May  1 14:56 core.38575
drwx------   2 root root  16384 Jun 25  2025 lost+found
drwxr-xr-x 207 root root  12288 Jun  8 15:18 etc
drwxr-xr-x   4 root root   4096 May 27 20:43 boot
drwxr-xr-x   3 root root   4096 Jun 25  2025 home
drwxr-xr-x   5 root root   4096 Jun 26  2025 media
drwxr-xr-x   2 root root   4096 Jun 25  2025 mnt
drwxr-xr-x   6 root root   4096 Jun  8 14:57 opt
drwx------  10 root root   4096 Jun  8 15:11 root
drwxr-xr-x   3 root root   4096 Jun 26  2025 srv
drwxrwxr-x   9 root root   4096 Jun 26  2025 timeshift
drwxr-xr-x  16 root root   4096 Jul 17  2025 usr
drwxr-xr-x  12 root root   4096 Jul 17  2025 var
drwxr-xr-x  20 root root   3800 Jun  8 15:18 dev
drwxr-xr-x  44 root root   1020 Jun  8 15:18 run
drwxrwxrwt  15 root root    380 Jun  8 15:43 tmp
lrwxrwxrwx   1 root root     35 Jul  2  2025 initrd.img.old -> boot/initrd.img-6.12.33+deb13-amd64
lrwxrwxrwx   1 root root     34 Jul  2  2025 initrd.img -> boot/initrd.img-6.12.33+kali-amd64
lrwxrwxrwx   1 root root     32 Jul  2  2025 vmlinuz.old -> boot/vmlinuz-6.12.33+deb13-amd64
lrwxrwxrwx   1 root root     31 Jul  2  2025 vmlinuz -> boot/vmlinuz-6.12.33+kali-amd64
lrwxrwxrwx   1 root root      9 Jun 26  2025 lib32 -> usr/lib32
lrwxrwxrwx   1 root root      9 Jun 25  2025 lib64 -> usr/lib64
lrwxrwxrwx   1 root root      8 Jun 25  2025 sbin -> usr/sbin
lrwxrwxrwx   1 root root      7 Jun 25  2025 bin -> usr/bin
lrwxrwxrwx   1 root root      7 Jun 25  2025 lib -> usr/lib
dr-xr-xr-x 446 root root      0 Jun  8  2026 proc
dr-xr-xr-x  13 root root      0 Jun  8 15:48 sys
```

3. List directory contents along with their inode no. (man ls)

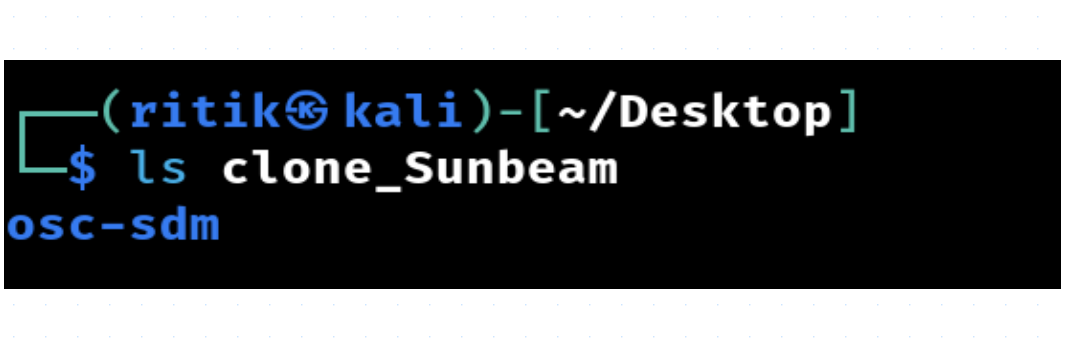
ls -i
ls -li
ls -ail
ls -i *
ls -lih



```
┌──(ritik㉿kali)-[/]
└─$ ls -i
     13 bin          524289 home               15 lib64          1 proc        1 sys            17 vmlinuz
5373953 boot             18 initrd.img         11 lost+found  7208961 root  7602177 timeshift      19 vmlinuz.old
     22 core.38575       20 initrd.img.old  6291457 media          1 run         1 tmp
      1 dev              14 lib             1572865 mnt           16 sbin  4980737 usr
7077889 etc              21 lib32           7340033 opt      1179649 srv   6684673 var
```

4. List the contents of the subdirectory.

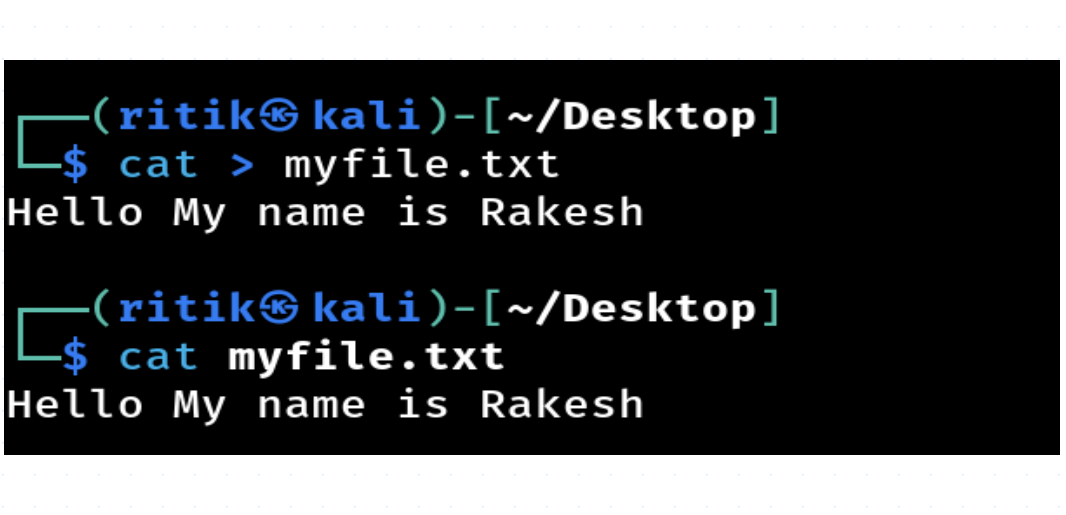
ls subdir
ls -l subdir
ls -a subdir
ls -R subdir
ls -lah subdir



```
┌──(ritik㉿kali)-[~/Desktop]
└─$ ls clone_Sunbeam
osc-sdm
```

5. Create a file, write your surname and name in it and save.

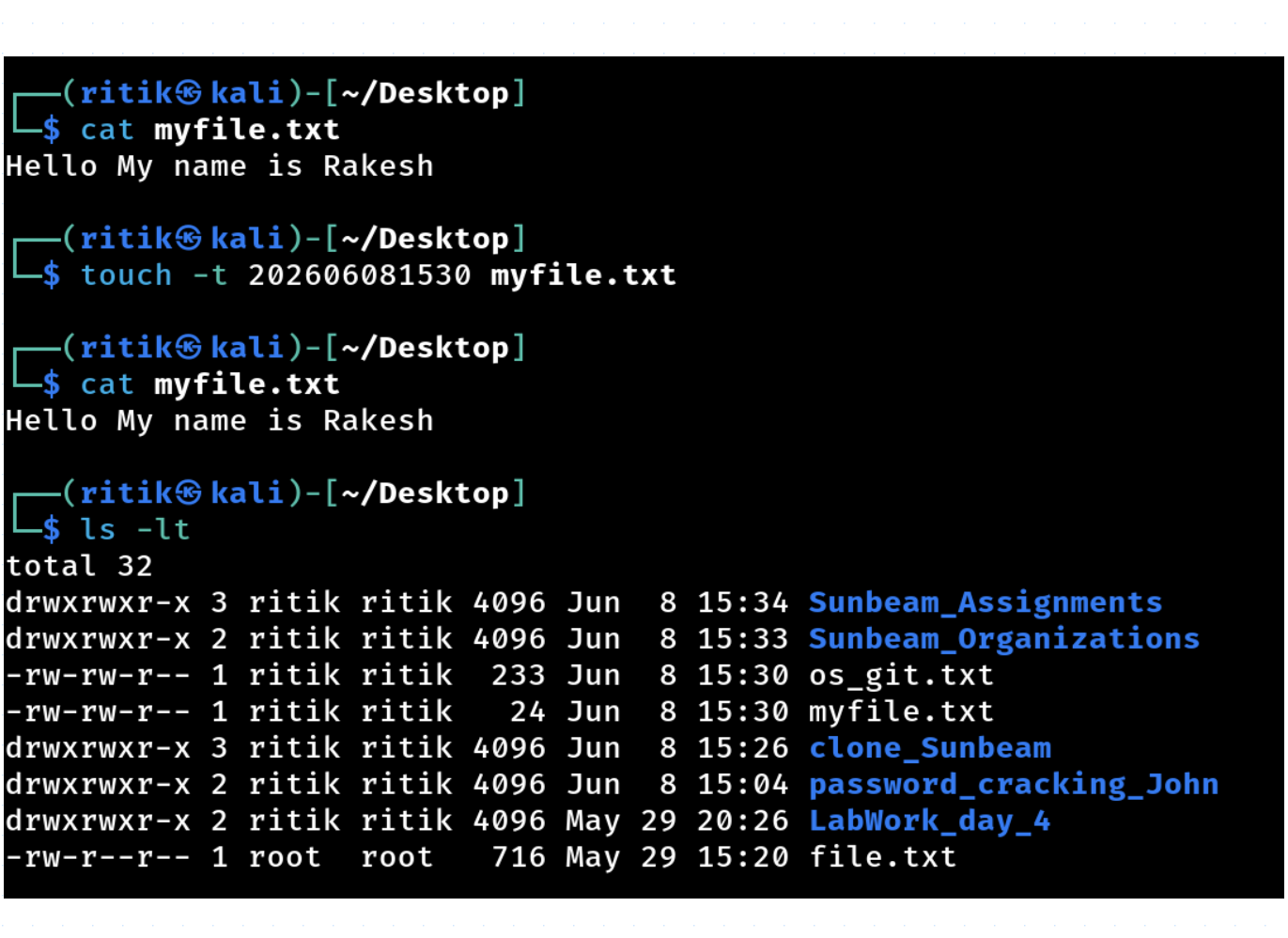
i. now open the same file and add your address in it and save.
ii. reopen the same file and check your name and address in it.



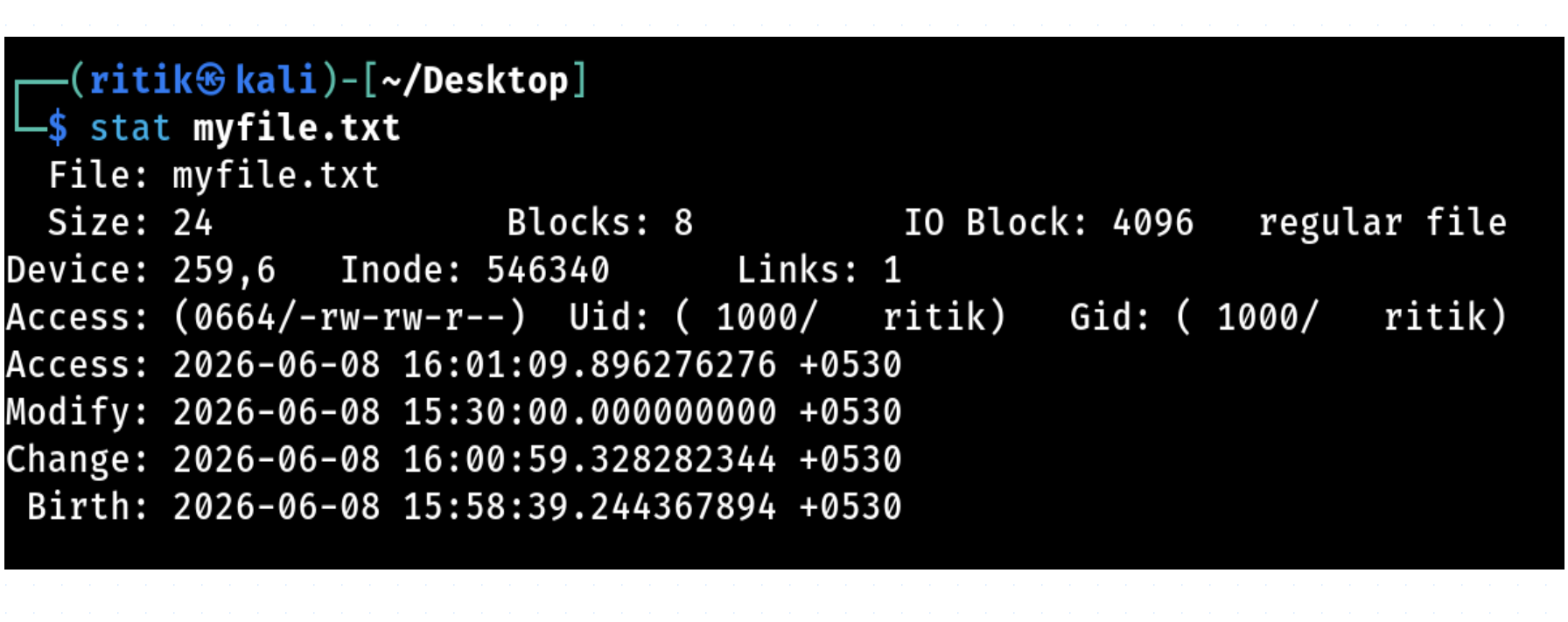
```
┌──(ritik㉿kali)-[~/Desktop]
└─$ cat > myfile.txt
Hello My name is Rakesh

┌──(ritik㉿kali)-[~/Desktop]
└─$ cat myfile.txt
Hello My name is Rakesh
```

6. how to change the time stamp of file (man touch)

touch myfile.txt
touch -a myfile.txt
touch -m myfile.txt
touch -t 202606081200 myfile.txt
touch -r file1 file2



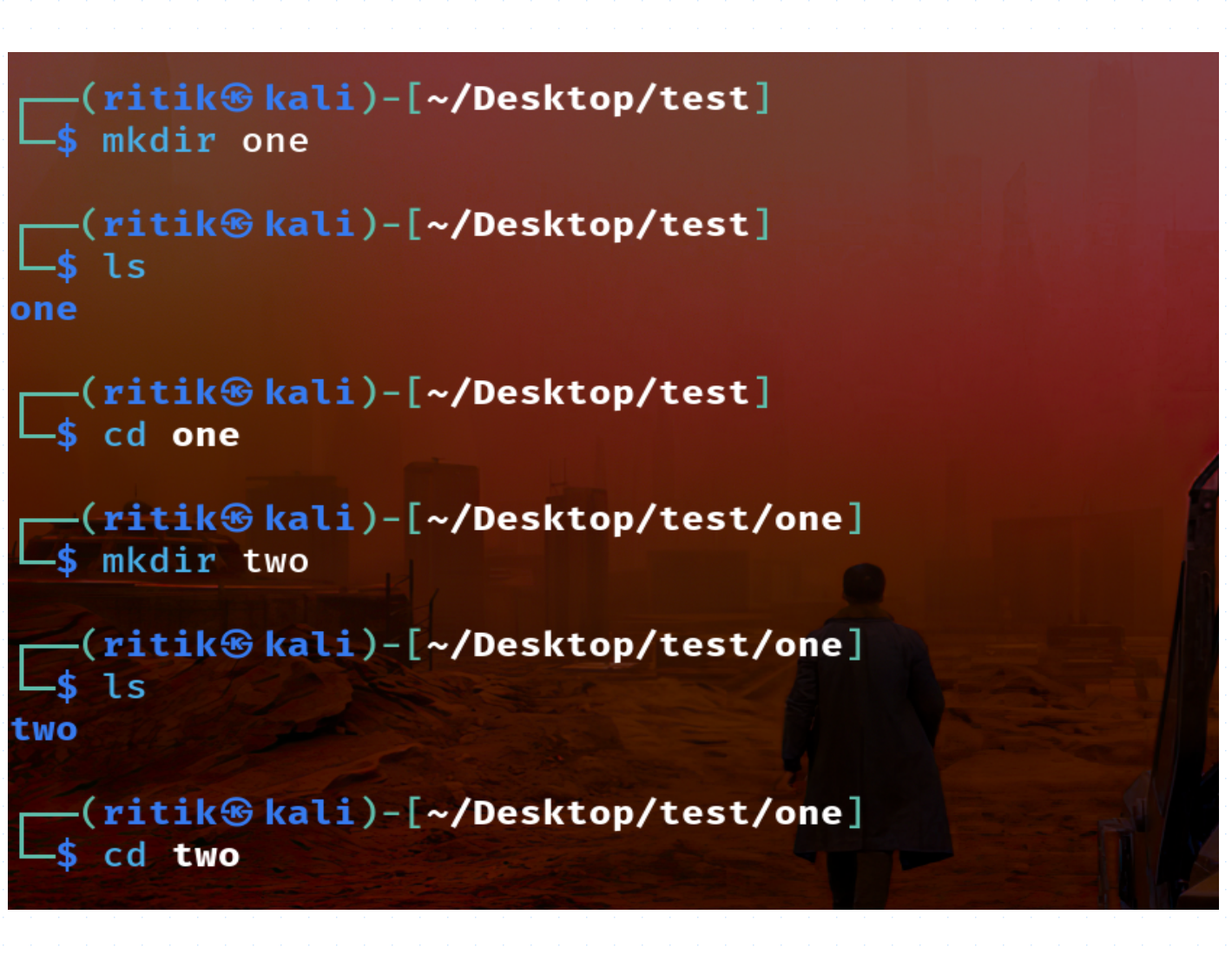
```
┌──(ritik㉿kali)-[~/Desktop]
└─$ cat myfile.txt
Hello My name is Rakesh

┌──(ritik㉿kali)-[~/Desktop]
└─$ touch -t 202606081530 myfile.txt

┌──(ritik㉿kali)-[~/Desktop]
└─$ cat myfile.txt
Hello My name is Rakesh

┌──(ritik㉿kali)-[~/Desktop]
└─$ ls -lt
total 32
drwxrwxr-x 3 ritik ritik 4096 Jun  8 15:34 Sunbeam_Assignments
drwxrwxr-x 2 ritik ritik 4096 Jun  8 15:33 Sunbeam_Organizations
-rw-rw-r-- 1 ritik ritik  233 Jun  8 15:30 os_git.txt
-rw-rw-r-- 1 ritik ritik   24 Jun  8 15:30 myfile.txt
drwxrwxr-x 3 ritik ritik 4096 Jun  8 15:26 clone_Sunbeam
drwxrwxr-x 2 ritik ritik 4096 Jun  8 15:04 password_cracking_John
drwxrwxr-x 2 ritik ritik 4096 May 29 20:26 LabWork_day_4
-rw-r--r-- 1 root  root   716 May 29 15:20 file.txt
```



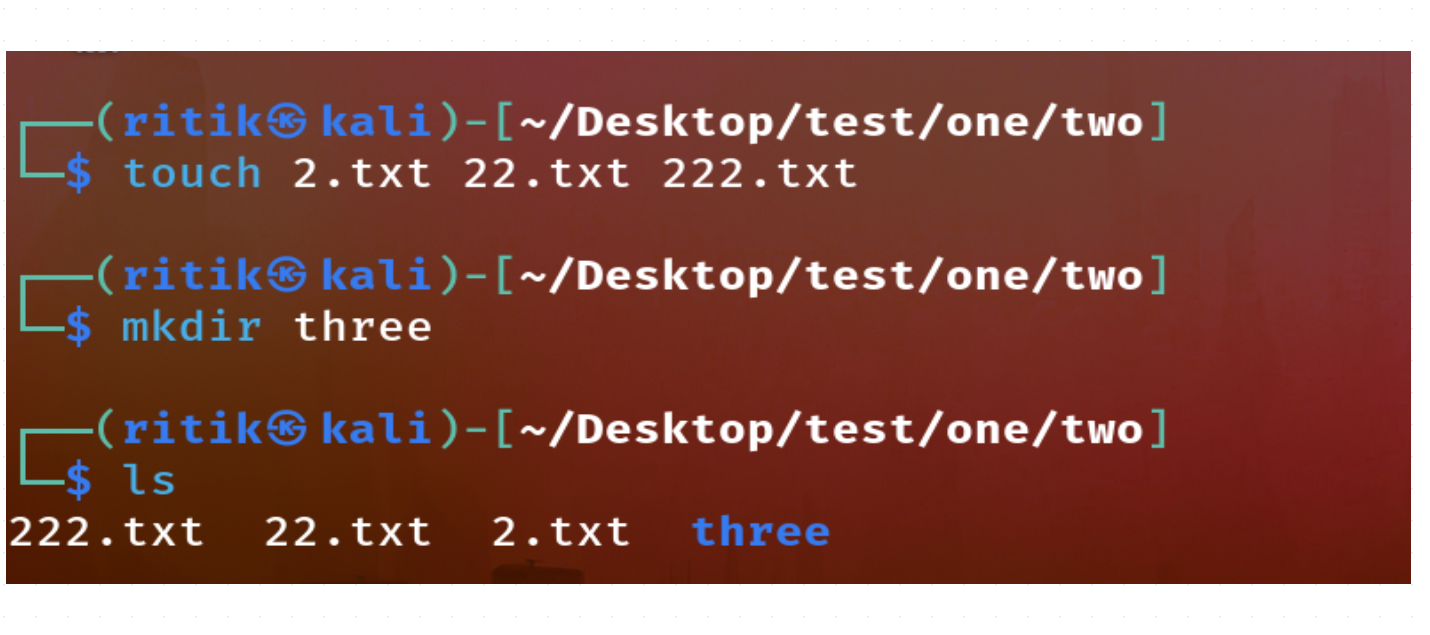
```
┌──(ritik㉿kali)-[~/Desktop]
└─$ stat myfile.txt
  File: myfile.txt
  Size: 24              Blocks: 8          IO Block: 4096   regular file
Device: 259,6   Inode: 546340      Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/   ritik)   Gid: ( 1000/   ritik)
Access: 2026-06-08 16:01:09.896276276 +0530
Modify: 2026-06-08 15:30:00.000000000 +0530
Change: 2026-06-08 16:00:59.328282344 +0530
 Birth: 2026-06-08 15:58:39.244367894 +0530
```

7. Create directory structure : Give following commands ,

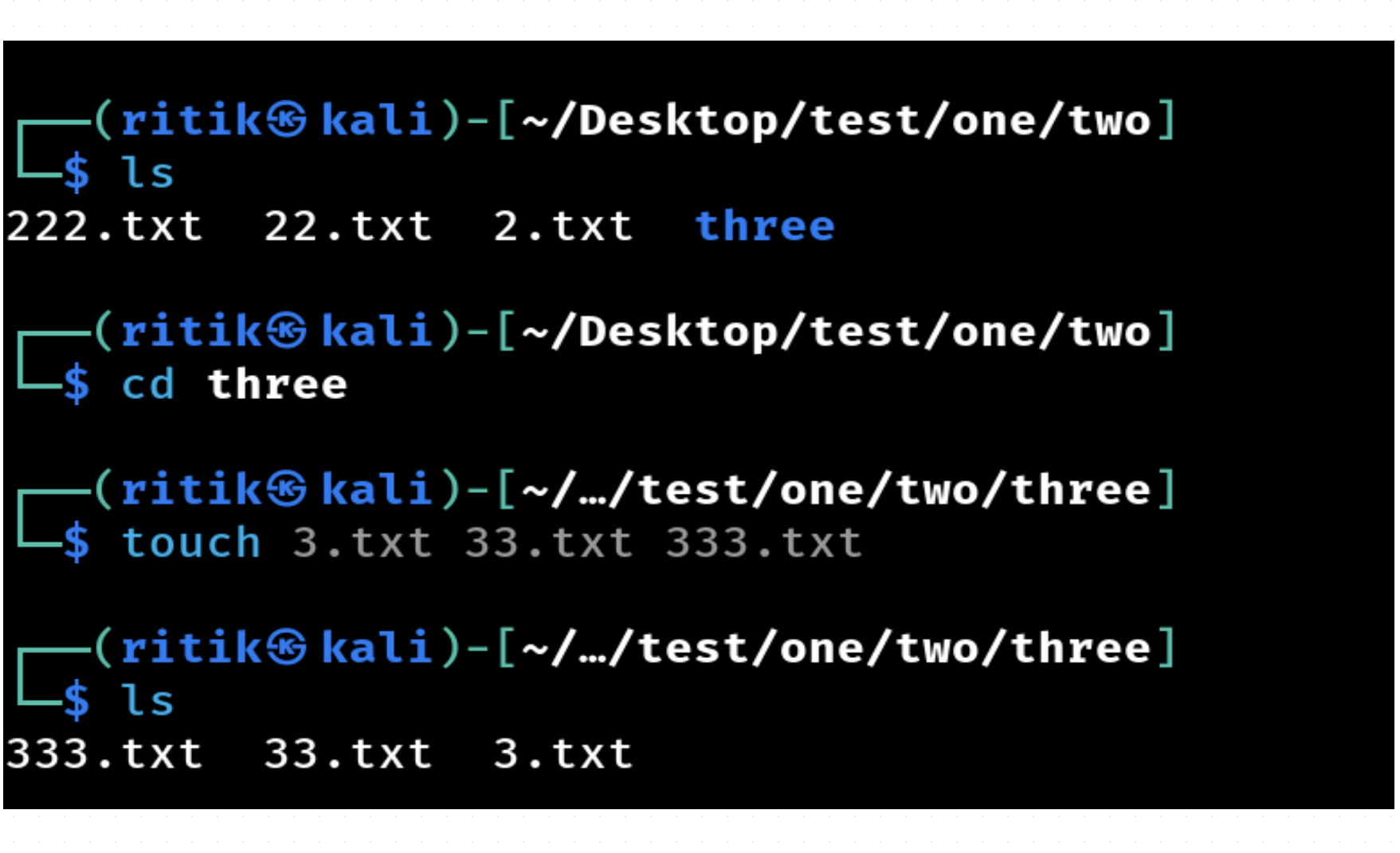
```
    -> mkdir one
    -> cd one
    -> touch 1.txt 11.txt 111.txt
    -> mkdir two
    -> cd two
    -> touch 2.txt 22.txt 222.txt
    -> mkdir three
    -> cd three
    -> touch 3.txt 33.txt 333.txt
    -> mkdir four
    -> cd four
    -> touch 4.txt 44.txt 444.txt
    -> mkdir five
    -> cd five
    -> touch 5.txt 55.txt 555.txt
    -> cd ~   (i.e. go to your home directory)
```



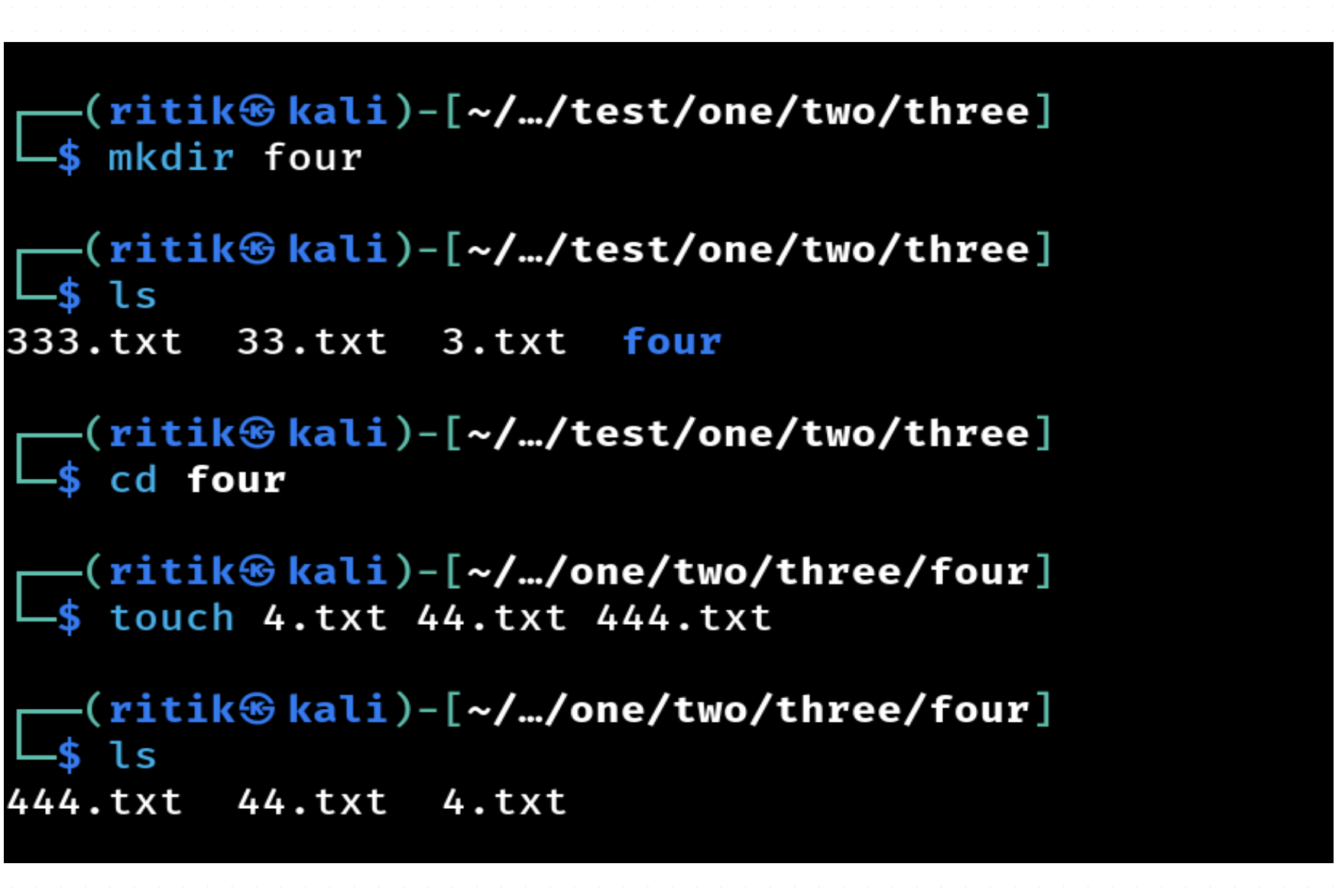
```
┌──(ritik㉿kali)-[~/Desktop/test]
└─$ mkdir one

┌──(ritik㉿kali)-[~/Desktop/test]
└─$ ls
one

┌──(ritik㉿kali)-[~/Desktop/test]
└─$ cd one

┌──(ritik㉿kali)-[~/Desktop/test/one]
└─$ mkdir two

┌──(ritik㉿kali)-[~/Desktop/test/one]
└─$ ls
two

┌──(ritik㉿kali)-[~/Desktop/test/one]
└─$ cd two
```



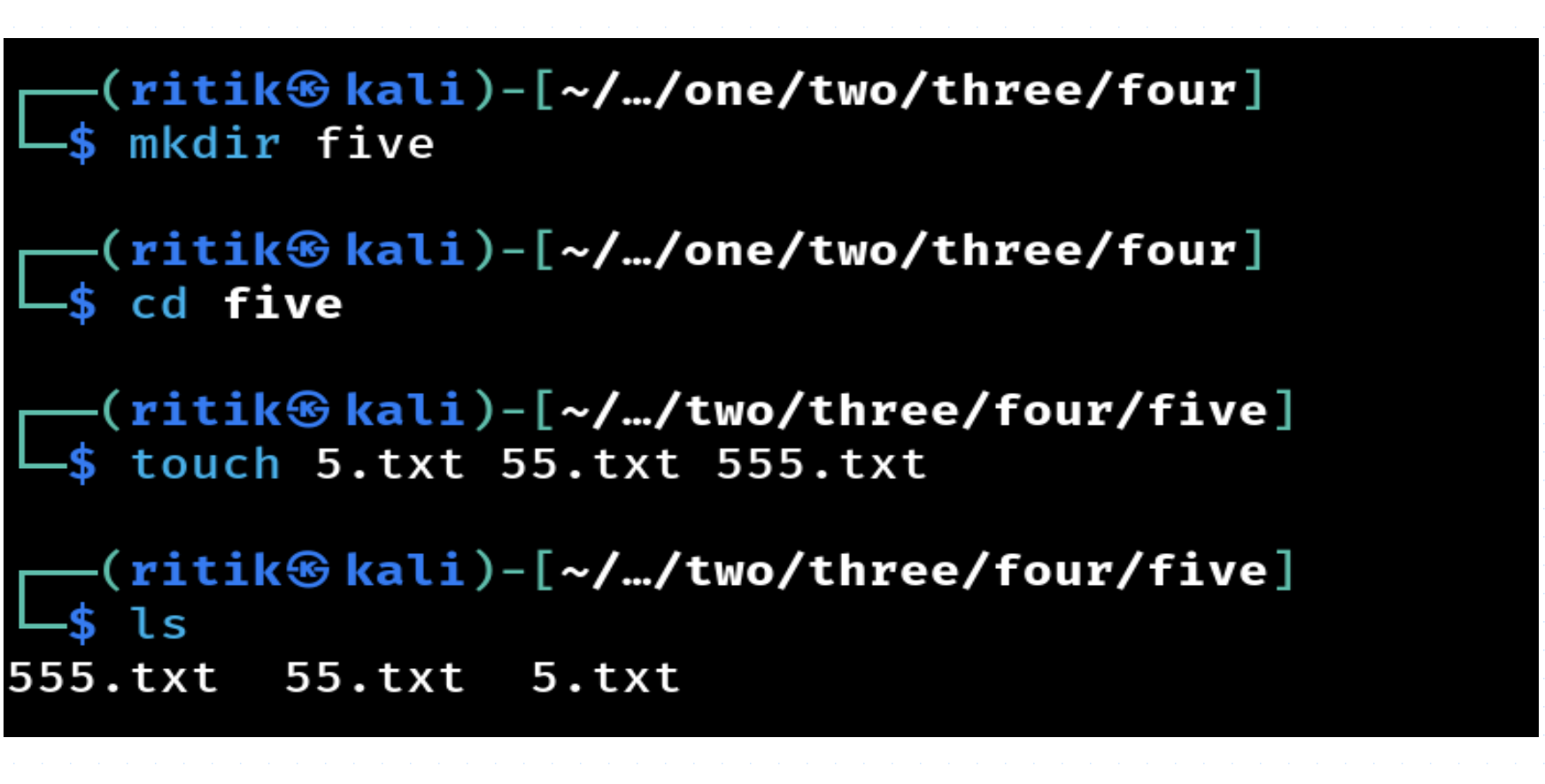
```
┌──(ritik㉿kali)-[~/Desktop/test/one/two]
└─$ touch 2.txt 22.txt 222.txt

┌──(ritik㉿kali)-[~/Desktop/test/one/two]
└─$ mkdir three

┌──(ritik㉿kali)-[~/Desktop/test/one/two]
└─$ ls
222.txt  22.txt  2.txt  three
```



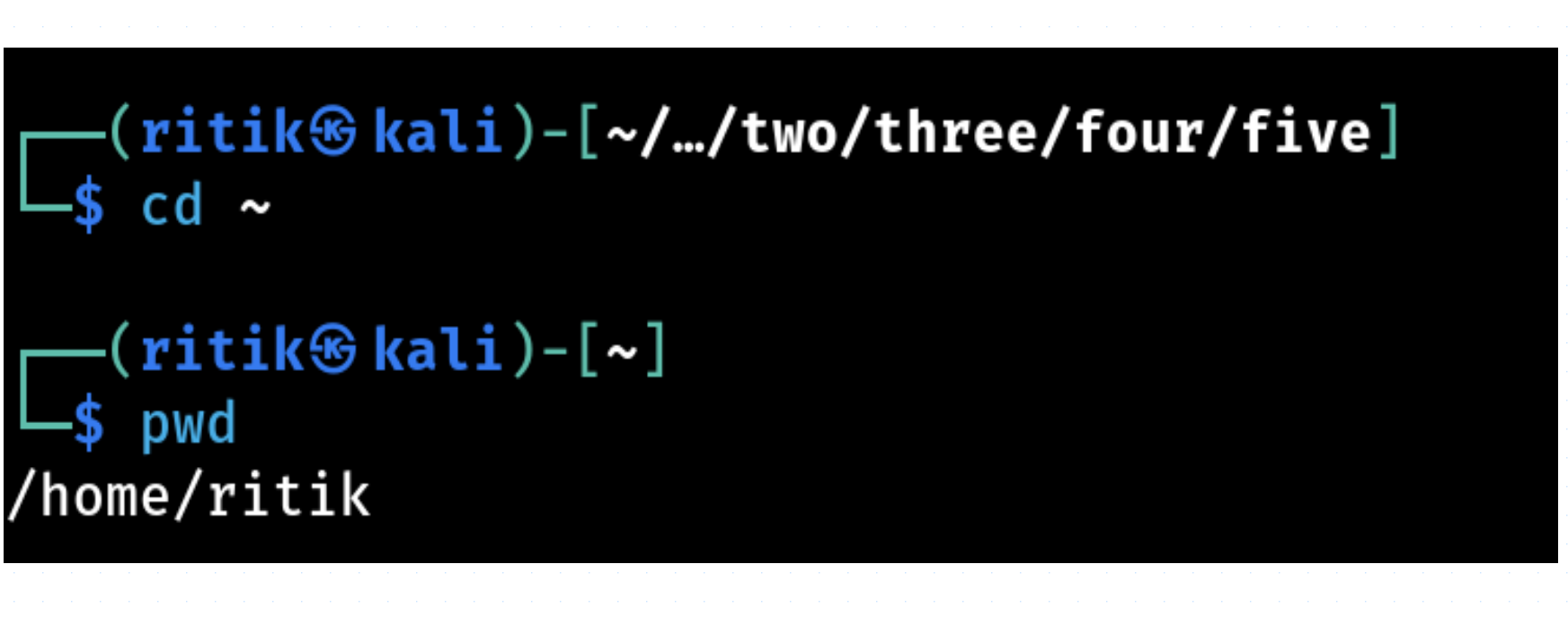
```
┌──(ritik㉿kali)-[~/Desktop/test/one/two]
└─$ ls
222.txt  22.txt  2.txt  three

┌──(ritik㉿kali)-[~/Desktop/test/one/two]
└─$ cd three

┌──(ritik㉿kali)-[~/…/test/one/two/three]
└─$ touch 3.txt 33.txt 333.txt

┌──(ritik㉿kali)-[~/…/test/one/two/three]
└─$ ls
333.txt  33.txt  3.txt
```



```
┌──(ritik㉿kali)-[~/…/test/one/two/three]
└─$ mkdir four

┌──(ritik㉿kali)-[~/…/test/one/two/three]
└─$ ls
333.txt  33.txt  3.txt  four

┌──(ritik㉿kali)-[~/…/test/one/two/three]
└─$ cd four

┌──(ritik㉿kali)-[~/…/one/two/three/four]
└─$ touch 4.txt 44.txt 444.txt

┌──(ritik㉿kali)-[~/…/one/two/three/four]
└─$ ls
444.txt  44.txt  4.txt
```



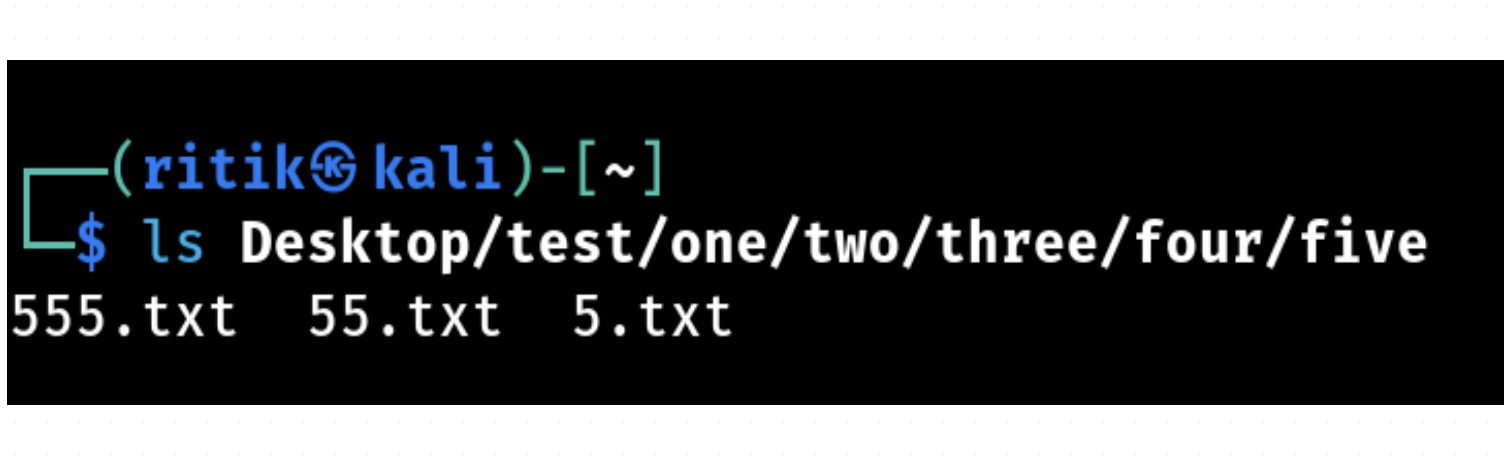
```
┌──(ritik㉿kali)-[~/…/one/two/three/four]
└─$ mkdir five

┌──(ritik㉿kali)-[~/…/one/two/three/four]
└─$ cd five

┌──(ritik㉿kali)-[~/…/two/three/four/five]
└─$ touch 5.txt 55.txt 555.txt

┌──(ritik㉿kali)-[~/…/two/three/four/five]
└─$ ls
555.txt  55.txt  5.txt
```



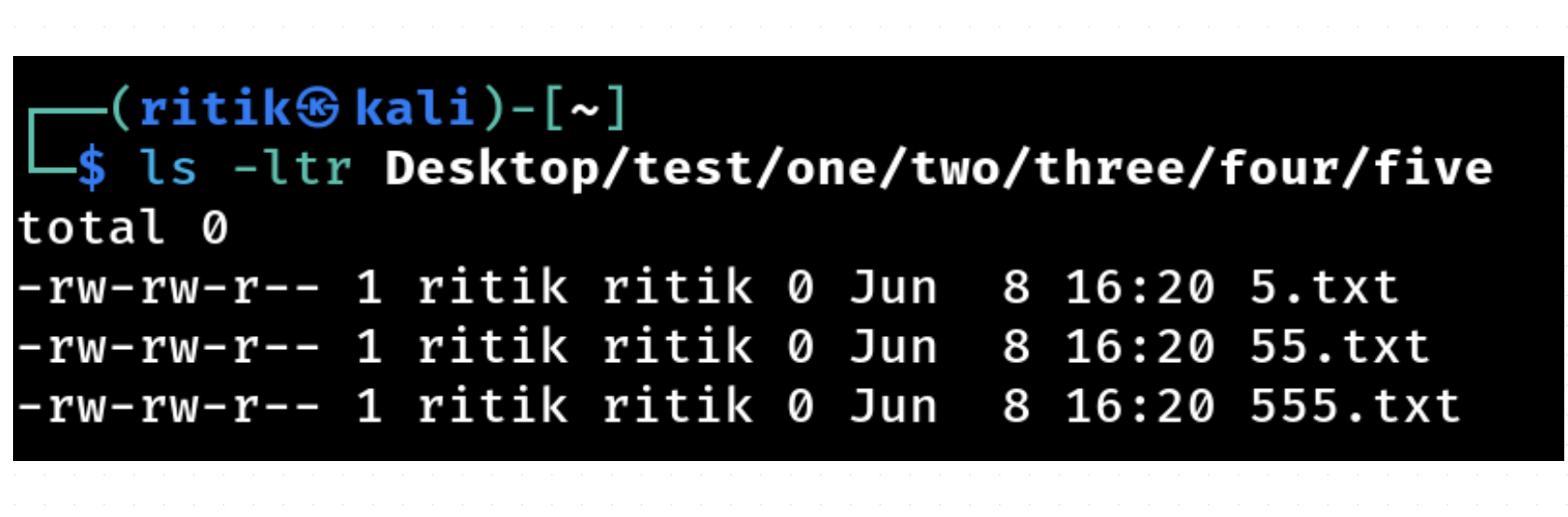
```
┌──(ritik㉿kali)-[~/…/two/three/four/five]
└─$ cd ~

┌──(ritik㉿kali)-[~]
└─$ pwd
/home/ritik
```

8. Considering above directory structure , Do following operations :

i. list the directory contents of directory named "five" from current directory (i.e. home directory).

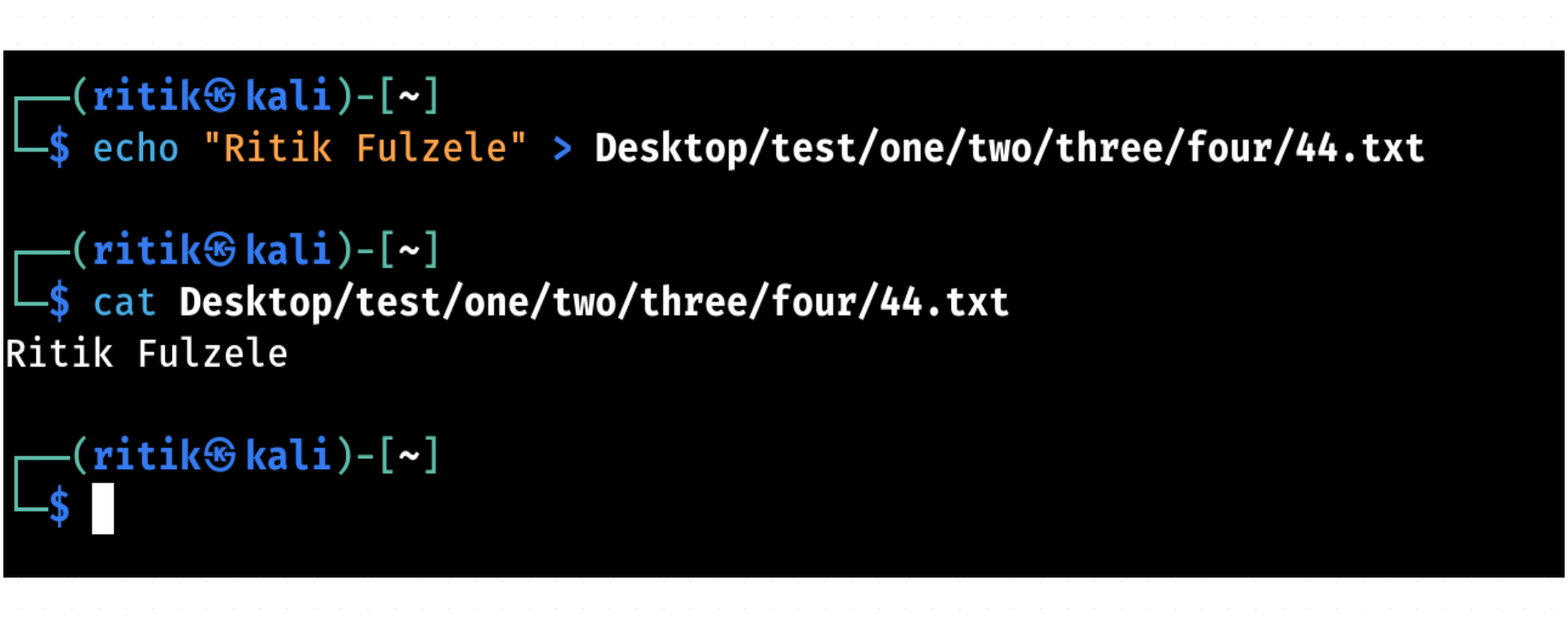


```
┌──(ritik㉿kali)-[~]
└─$ ls Desktop/test/one/two/three/four/five
555.txt  55.txt  5.txt
```



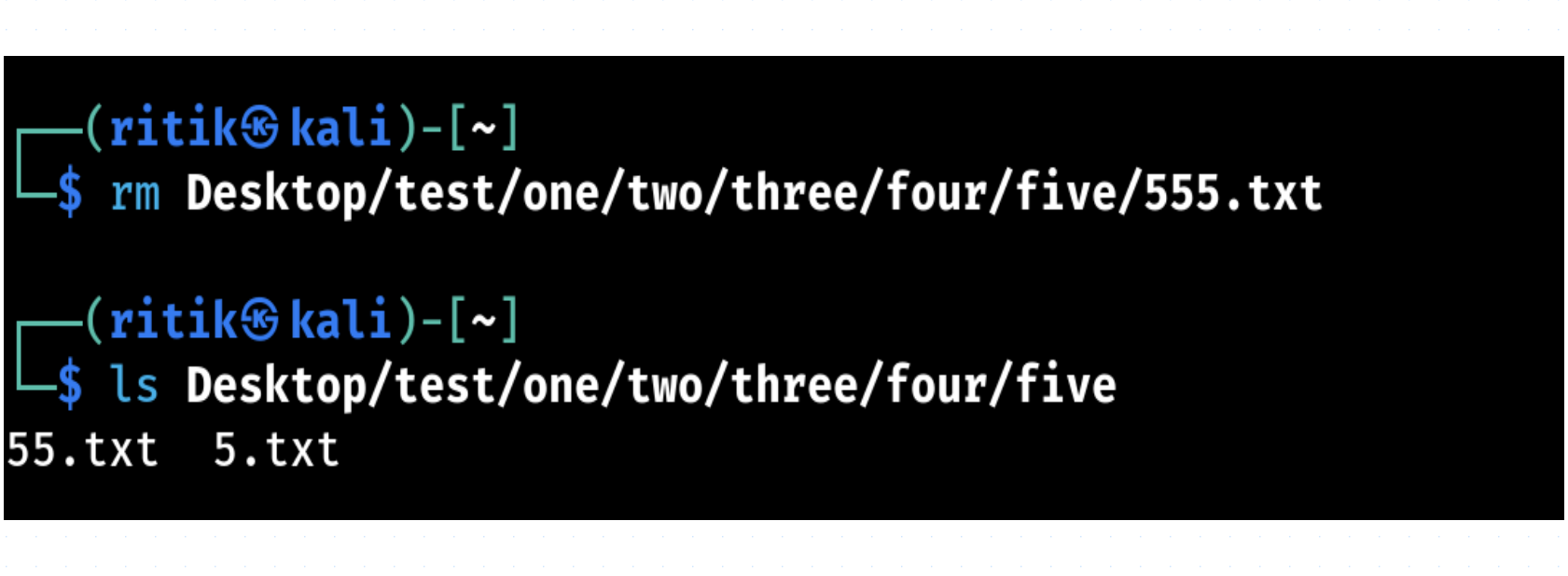
```
┌──(ritik㉿kali)-[~]
└─$ ls -ltr Desktop/test/one/two/three/four/five
total 0
-rw-rw-r-- 1 ritik ritik 0 Jun  8 16:20 5.txt
-rw-rw-r-- 1 ritik ritik 0 Jun  8 16:20 55.txt
-rw-rw-r-- 1 ritik ritik 0 Jun  8 16:20 555.txt
```

8(ii). Write your name in file "44.txt" of directory four from home directory



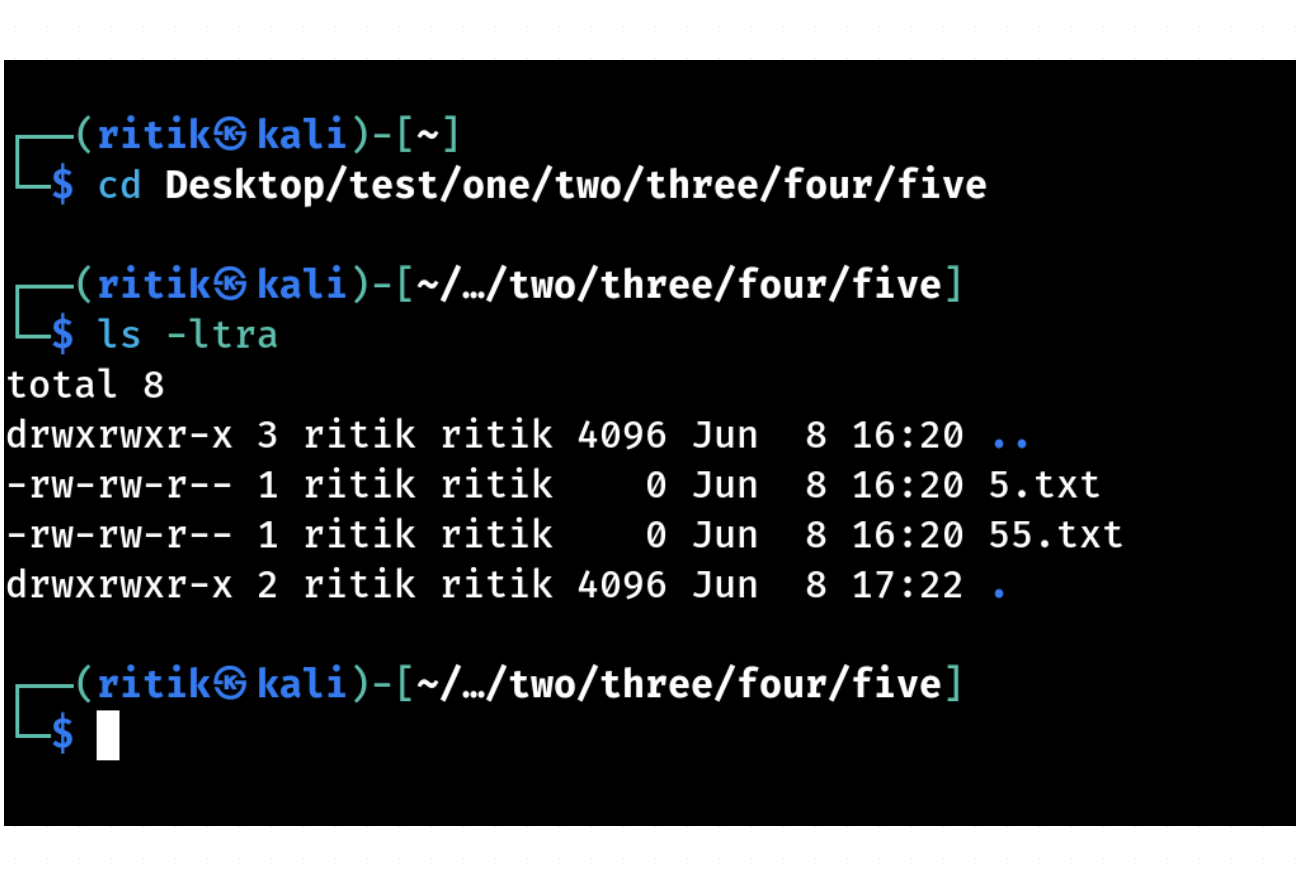
```
┌──(ritik㉿kali)-[~]
└─$ echo "Ritik Fulzele" > Desktop/test/one/two/three/four/44.txt

┌──(ritik㉿kali)-[~]
└─$ cat Desktop/test/one/two/three/four/44.txt
Ritik Fulzele

┌──(ritik㉿kali)-[~]
└─$ █
```

8(iii). Remove file "555.txt" from directory five from home directory



```
┌──(ritik㉿kali)-[~]
└─$ rm Desktop/test/one/two/three/four/five/555.txt

┌──(ritik㉿kali)-[~]
└─$ ls Desktop/test/one/two/three/four/five
55.txt  5.txt
```

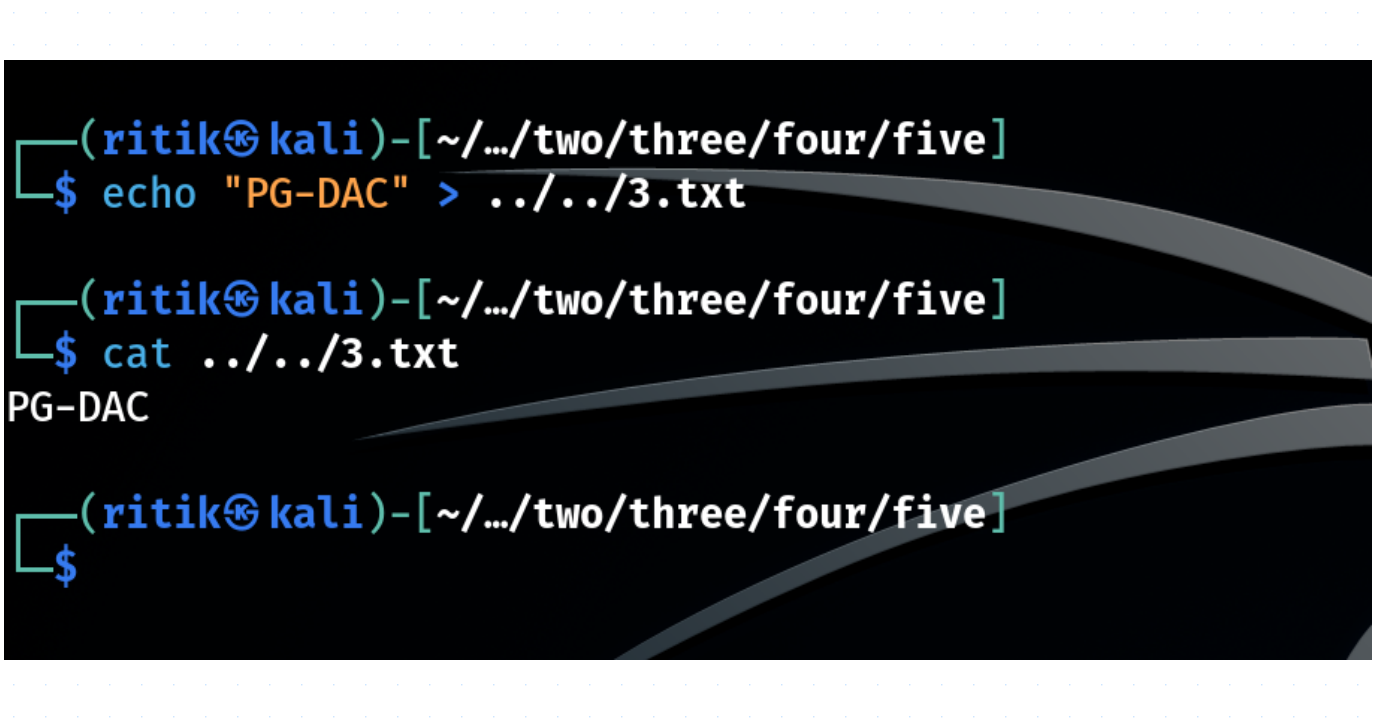
8(iv). Change directory to five



```
┌──(ritik㉿kali)-[~]
└─$ cd Desktop/test/one/two/three/four/five

┌──(ritik㉿kali)-[~/…/two/three/four/five]
└─$ ls -ltra
total 8
drwxrwxr-x 3 ritik ritik 4096 Jun  8 16:20 ..
-rw-rw-r-- 1 ritik ritik    0 Jun  8 16:20 5.txt
-rw-rw-r-- 1 ritik ritik    0 Jun  8 16:20 55.txt
drwxrwxr-x 2 ritik ritik 4096 Jun  8 17:22 .

┌──(ritik㉿kali)-[~/…/two/three/four/five]
└─$ █
```

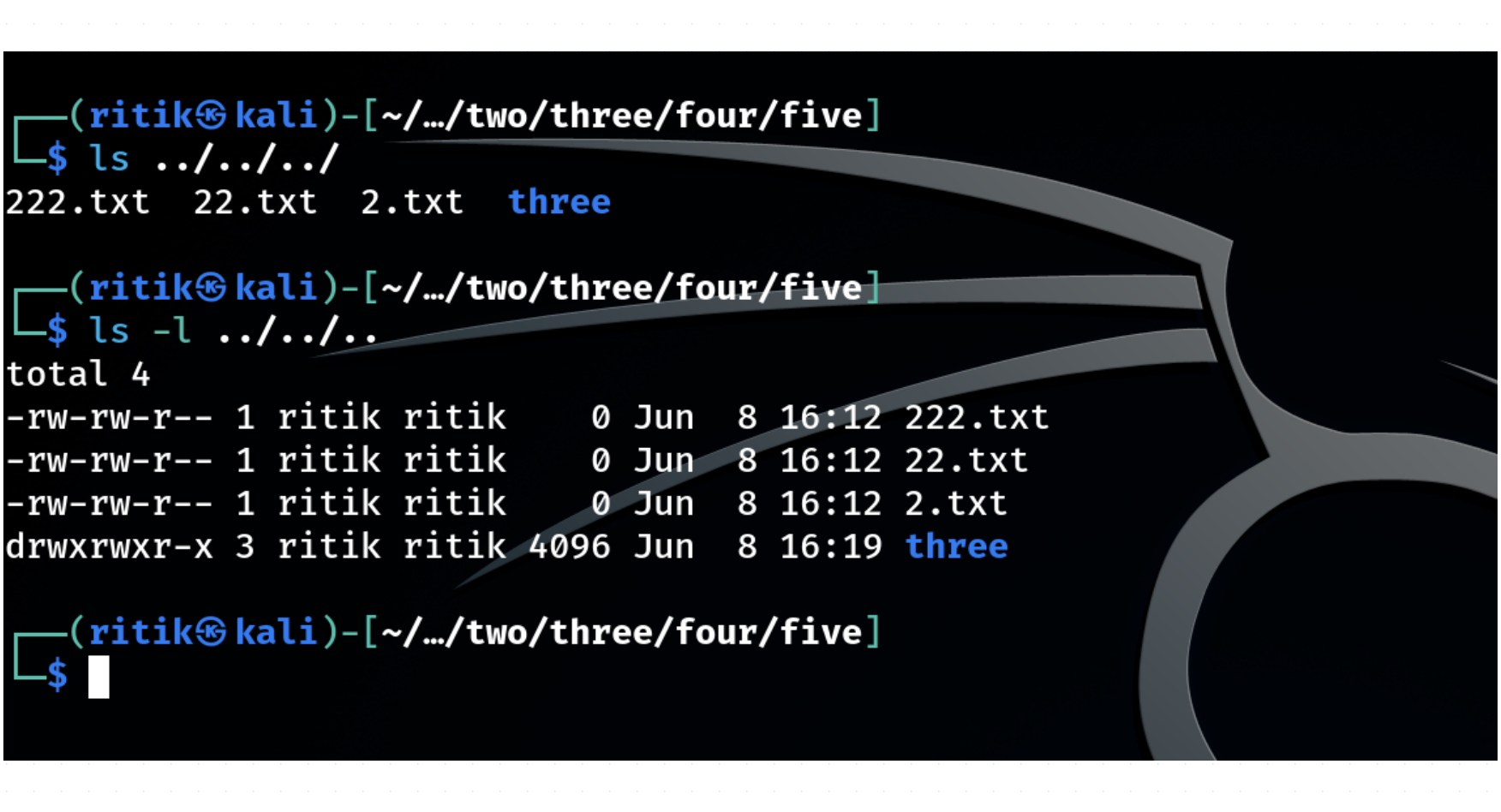
8(v). Write your course name in file "3.txt" from current directory (five)

echo "PG-DAC" > ../../3.txt



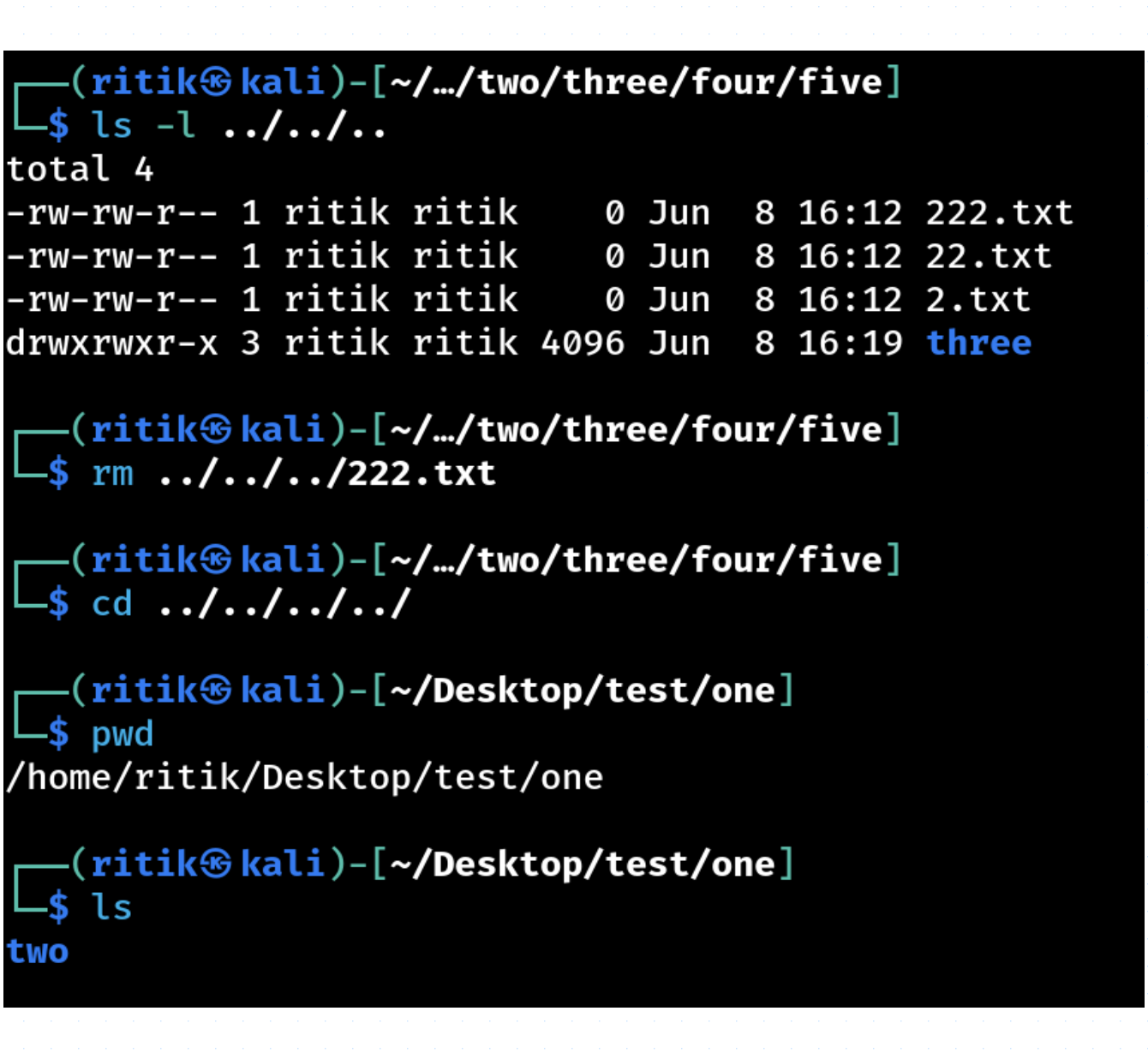
```
┌──(ritik㉿kali)-[~/…/two/three/four/five]
└─$ echo "PG-DAC" > ../../3.txt

┌──(ritik㉿kali)-[~/…/two/three/four/five]
└─$ cat ../../3.txt
PG-DAC

┌──(ritik㉿kali)-[~/…/two/three/four/five]
└─$
```

8(vi). List contents of directory "two" from current directory (five)



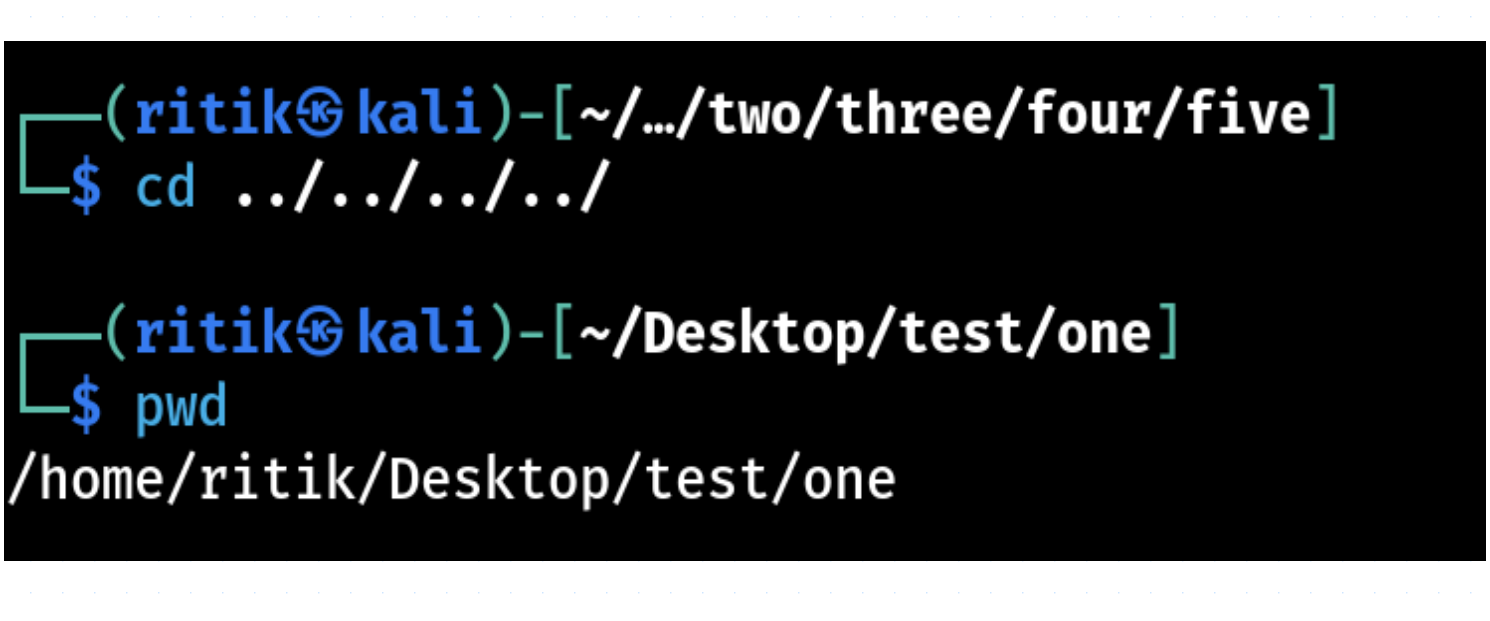
```
┌──(ritik㉿kali)-[~/…/two/three/four/five]
└─$ ls ../../../
222.txt  22.txt  2.txt  three

┌──(ritik㉿kali)-[~/…/two/three/four/five]
└─$ ls -l ../../..
total 4
-rw-rw-r-- 1 ritik ritik    0 Jun  8 16:12 222.txt
-rw-rw-r-- 1 ritik ritik    0 Jun  8 16:12 22.txt
-rw-rw-r-- 1 ritik ritik    0 Jun  8 16:12 2.txt
drwxrwxr-x 3 ritik ritik 4096 Jun  8 16:19 three

┌──(ritik㉿kali)-[~/…/two/three/four/five]
└─$ █
```

8(vii). Remove file "222.txt" from directory two from current directory (five)



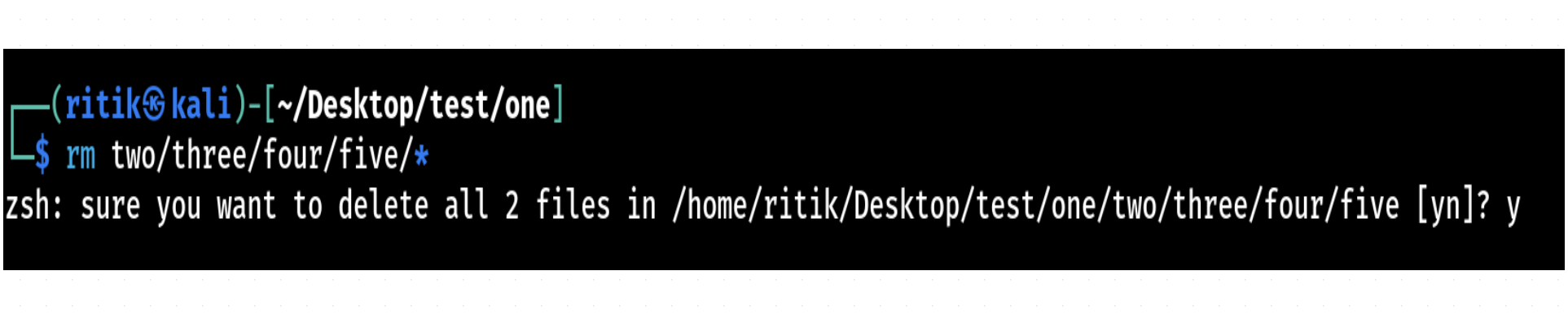
```
┌──(ritik㉿kali)-[~/…/two/three/four/five]
└─$ ls -l ../../..
total 4
-rw-rw-r-- 1 ritik ritik    0 Jun  8 16:12 222.txt
-rw-rw-r-- 1 ritik ritik    0 Jun  8 16:12 22.txt
-rw-rw-r-- 1 ritik ritik    0 Jun  8 16:12 2.txt
drwxrwxr-x 3 ritik ritik 4096 Jun  8 16:19 three

┌──(ritik㉿kali)-[~/…/two/three/four/five]
└─$ rm ../../../222.txt

┌──(ritik㉿kali)-[~/…/two/three/four/five]
└─$ cd ../../../../

┌──(ritik㉿kali)-[~/Desktop/test/one]
└─$ pwd
/home/ritik/Desktop/test/one

┌──(ritik㉿kali)-[~/Desktop/test/one]
└─$ ls
two
```

8(viii). Change directory to "one"



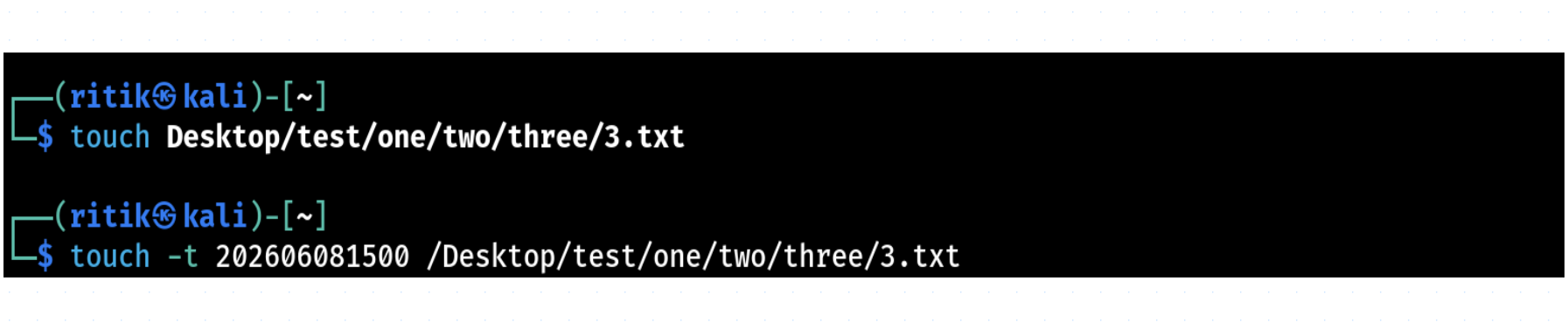
```
┌──(ritik㉿kali)-[~/…/two/three/four/five]
└─$ cd ../../../../

┌──(ritik㉿kali)-[~/Desktop/test/one]
└─$ pwd
/home/ritik/Desktop/test/one
```

8(ix). Remove all files only from directory "five" from current directory (one)



```
┌──(ritik㉿kali)-[~/Desktop/test/one]
└─$ rm two/three/four/five/*
zsh: sure you want to delete all 2 files in /home/ritik/Desktop/test/one/two/three/four/five [yn]? y
```

9. Change timestamp of file "3.txt" in directory "three"



```
┌──(ritik㉿kali)-[~]
└─$ touch Desktop/test/one/two/three/3.txt

┌──(ritik㉿kali)-[~]
└─$ touch -t 202606081500 /Desktop/test/one/two/three/3.txt
```